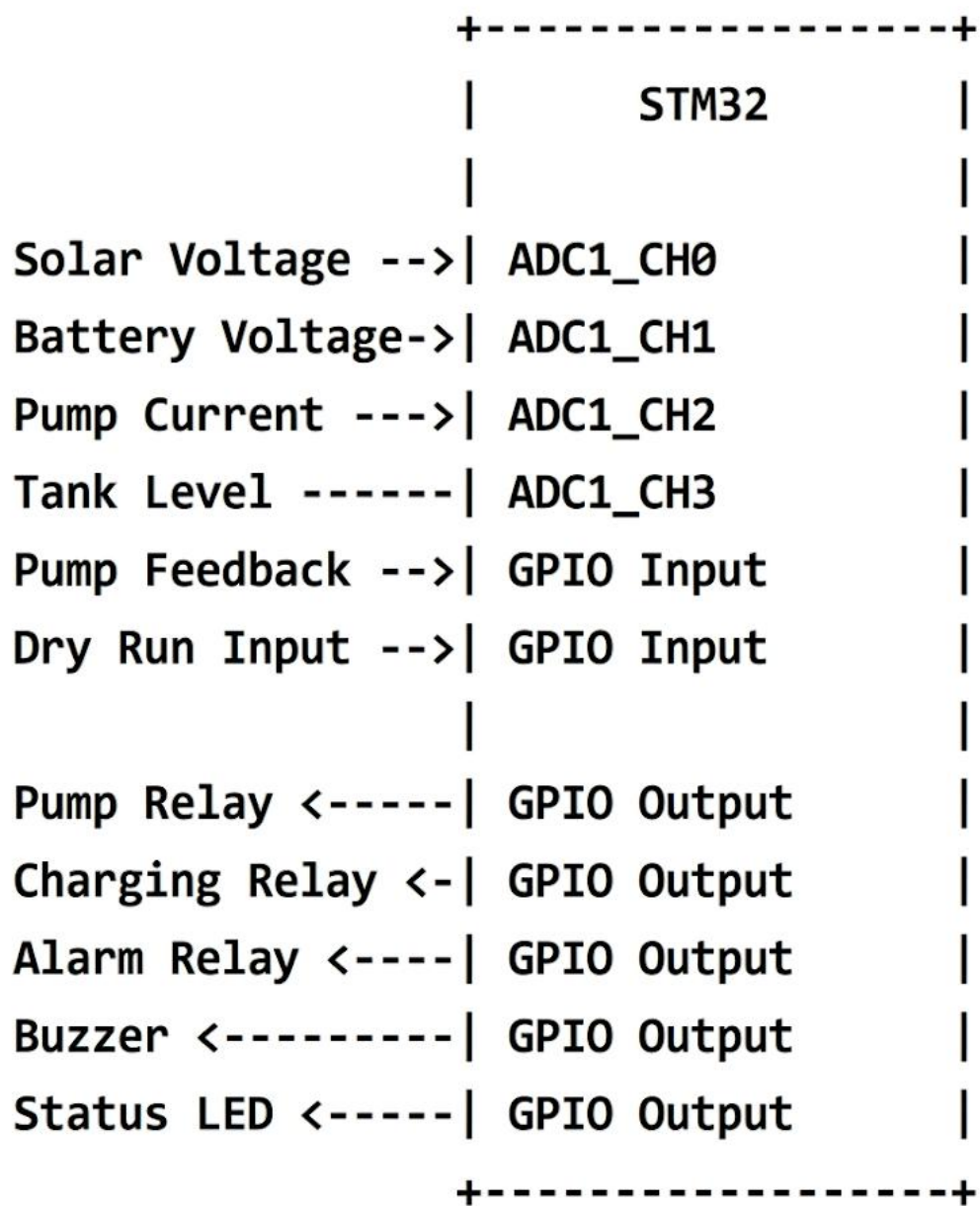


Task 9 – Solar Pump Monitoring Controller

STM32 Requirement

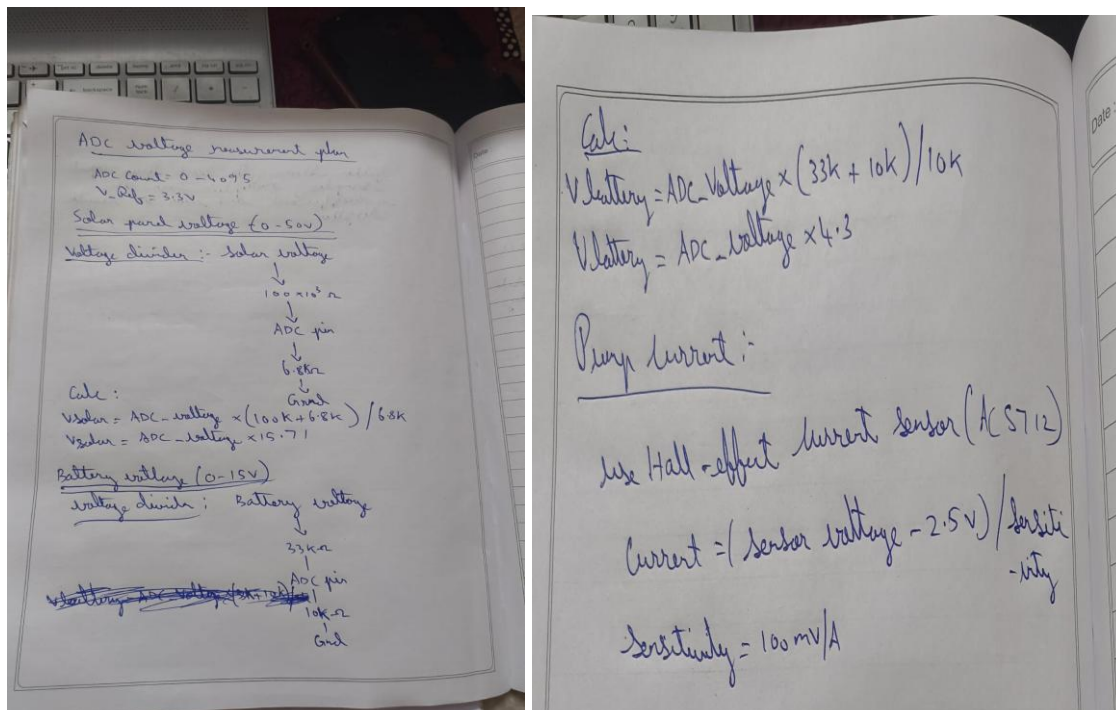
Block Diagram



GPIO allocation table

Function	STM32 Pin	Type
Solar Panel Voltage	PA0 (ADC1_CH0)	Analog Input
Battery Voltage	PA1 (ADC1_CH1)	Analog Input
Pump Current	PA2 (ADC1_CH2)	Analog Input
Tank Level Sensor	PA3 (ADC1_CH3)	Analog Input
Pump Feedback	PB0	Digital Input
Dry Run Input	PB1	Digital Input
Pump Relay	PB10	Digital Output
Charging Relay	PB11	Digital Output
Alarm Relay	PB12	Digital Output
Buzzer	PB13	Digital Output
Status LED	PC13	Digital Output

ADC voltage measurement plan



Battery protection logic

→Overcharge Protection

If Battery Voltage > 14.4V
Charging Relay OFF

→Recharge Enable

If Battery Voltage < 13.8V
Charging Relay ON

→Deep Discharge Protection

If Battery Voltage < 11.0V
Pump OFF
Alarm ON

→Recovery

If Battery Voltage > 12.0V
Alarm OFF
Pump Operation Allowed

Pump Control Logic

→Start Conditions

Battery Voltage > 12V
AND
Tank Level < 90%
AND
Dry Run = FALSE
AND
Pump Feedback OK

Then:

Pump Relay ON

Status LED ON

→Stop Conditions

Tank Full

Tank Level $\geq$ 90%

Pump Relay OFF

Dry Run Detected

Dry Run = TRUE

Pump Relay OFF

Alarm Relay ON

Buzzer ON

Low Battery

Battery Voltage < 11V

Pump Relay OFF

Alarm Relay ON

Pump Failure

Pump Relay ON

BUT

Pump Feedback = OFF for >5s

Pump Relay OFF

Alarm Relay ON

Buzzer ON